MAT1856/APM466 Assignment 1

Lu Huang, Student #: 1007636925

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Fundamental Questions - 25 points

1.
   (a) Governments issue bonds to fund spending without immediately expanding the money supply, because printing money can fuel inflation and undermine currency credibility.

   (b) A yield curve can flatten if investors expect weaker long-run growth/inflation (and future rate cuts), so long-term yields fall relative to short-term yields even as the central bank keeps short rates high.

   (c) Quantitative easing is when the Fed buys large amounts of securities to lower longer-term interest rates and support financial conditions; since COVID-19 began, it has used large-scale asset purchases to stabilize markets and provide accommodation.

2. The following 10 bonds were used (naming convention: "CAN *coupon month year*"):
   - CAN 0.25 Mar26 — ISIN: CA135087L518 — Coupon: 0.25
   - CAN 1 Sep26 — ISIN: CA135087L930 — Coupon: 1.00
   - CAN 1.25 Mar27 — ISIN: CA135087M847 — Coupon: 1.25
   - CAN 2.75 Sep27 — ISIN: CA135087N837 — Coupon: 2.75
   - CAN 3.5 Mar28 — ISIN: CA135087P576 — Coupon: 3.50
   - CAN 3.25 Sep28 — ISIN: CA135087Q491 — Coupon: 3.25
   - CAN 4 Mar29 — ISIN: CA135087Q988 — Coupon: 4.00
   - CAN 3.5 Sep29 — ISIN: CA135087R895 — Coupon: 3.50
   - CAN 2.75 Mar30 — ISIN: CA135087S471 — Coupon: 2.75
   - CAN 2.75 Sep30 — ISIN: CA135087T388 — Coupon: 2.75

   Together, these bonds provide ten observed yield points at different times to maturity within 0–5 years on each trading day. This maturity dispersion supports a stable daily YTM curve (constructed by interpolating across the observed points) and enables sequential bootstrapping of discount factors and spot rates over the same horizon.

3. The eigenvectors of the covariance matrix give the principal component directions, i.e., the dominant shapes of joint movements across points on the curve. For a yield curve these typically correspond to a level shift (roughly equal weights across maturities, like 1, 1, 1,..., a slope change (short and long maturities with opposite signs), and a curvature change (the middle moving opposite to the ends). The associated eigenvalues measure the variance captured by each component, so they tell us how much of the curve's total variation is explained by level, slope, curvature, etc.

Empirical Questions - 75 points

4.

(a) I used 1D piecewise linear interpolation to form a yield curve because yields are observed only at a few discrete maturities. With limited and unevenly spaced bond data, linear interpolation is stable, transparent, and avoids the overshoot/oscillations that can occur with splines. I also avoid extrapolation by only interpolating within the observed maturity range. (See Figure 1 in references)

(b) **Inputs:** For a fixed observation date t_0, bonds $i = 1, \dots, N$ with dirty price P_i, coupon rate c_i, maturity date M_i; face value F; frequency $m = 2$ (semiannual).

Helpers (brief):

- $\textsc{YearFrac}(t_0, d)$: ACT/365 year fraction from t_0 to cashflow date d.
- $\textsc{CouponSchedule}(t_0, M_i)$: remaining semiannual payment dates after t_0 up to M_i.

Step 1: Log-linear interpolation of discount factors. Maintain curve nodes $\{(T_k, \ln D_k)\}_{k=0}^{K}$ with $(T_0, \ln D_0) = (0, 0)$. For any $t > 0$:

$$\ln D(t) = \begin{cases} \ln D_k + \dfrac{t - T_k}{T_{k+1} - T_k}\big(\ln D_{k+1} - \ln D_k\big), & T_k \le t \le T_{k+1}, \\ \ln D_K + s\,(t - T_K), \quad s = \dfrac{\ln D_K - \ln D_{K-1}}{T_K - T_{K-1}}, & t > T_K, \end{cases} \qquad D(t) = e^{\ln D(t)}.$$

Step 2: Bootstrap for one observation date. Remove duplicate ISIN rows and sort bonds by increasing maturity time (TTM). Initialize nodes: $T_0 \leftarrow 0$, $\ln D_0 \leftarrow 0$.
for $i = 1, \dots, N$ **do:** Construct payment dates $\{d_{i,1}, \dots, d_{i,n_i}\} \leftarrow \textsc{CouponSchedule}(t_0, M_i)$.
Convert to times $t_{i,j} \leftarrow \textsc{YearFrac}(t_0, d_{i,j})$, and set $T_i \leftarrow t_{i,n_i}$. Coupon per period $C_i \leftarrow \dfrac{Fc_i}{m}$.
Compute PV of coupons before maturity using existing curve:

$$\mathrm{PV}_i^{\text{known}} = \sum_{j=1}^{n_i - 1} C_i\, D(t_{i,j}),$$

where $D(t_{i,j})$ is obtained by Step A from current nodes.
Solve for the new discount factor at maturity from the pricing identity:

$$P_i = \mathrm{PV}_i^{\text{known}} + (F + C_i)\, D(T_i) \quad \Longrightarrow \quad D(T_i) = \frac{P_i - \mathrm{PV}_i^{\text{known}}}{F + C_i}.$$

if $D(T_i) \le 0$ **then skip bond** i
Append node $(T_i, \ln D(T_i))$ to the curve.
Compute spot rate at T_i (continuous compounding):

$$r_{\text{cc}}(T_i) = -\frac{\ln D(T_i)}{T_i}.$$

semiannual spot:

$$r_{\text{sa}}(T_i) = m\left(D(T_i)^{-\frac{1}{mT_i}} - 1\right).$$

Output: bootstrapped points $\{(T_i, D(T_i), r_{\text{cc}}(T_i))\}$ for this observation date.

(c) **Inputs:** For each day, a bootstrapped curve table s with maturities T and either discount factors $D(T)$ or continuous spots $r_{cc}(T)$; starts $\mathcal{S} = \{1, 2, 3, 4\}$.

Helper (brief): Build nodes for interpolation $\textsc{BuildNodes}(s)$: keep rows with $T > 0$ and sort by T. **if** $D(T)$ *is available* **then**

set $D_k \leftarrow D(T_k)$

else

reconstruct $D_k \leftarrow \exp\big(-r_{cc}(T_k)\,T_k\big)$

prepend $(T_0, \ln D_0) = (0, 0)$ and form node arrays $\{(T_k, \ln D_k)\}_{k=0}^{K}$. drop duplicate T_k so that T_k is strictly increasing.

Helper: Log-linear interpolation of discount factors as defined in 4(b).

Procedure: Compute 1-year forwards for one day $(T_k, \ln D_k)_{k=0}^{K} \leftarrow \textsc{BuildNodes}(s)$. Initialize an empty output table $\mathcal{F}$.

foreach $T \in \mathcal{S}$ **do**

$D(T) \leftarrow \textsc{DFLogLinear}(T)$; $D(T+1) \leftarrow \textsc{DFLogLinear}(T+1)$. **if** $D(T) \leq 0$ ***or*** $D(T+1) \leq 0$ **then**

skip this start T

$$f_{cc}(T, T+1) \;=\; \ln\!\left(\frac{D(T)}{D(T+1)}\right)$$

Append (Start $= T$, End $= T+1$, Fwd_CC $= f_{cc}(T, T+1)$) to $\mathcal{F}$.

Main (across days): For each day, run the above procedure on its curve table s and store the resulting $\mathcal{F}$ (containing $1\to2, 2\to3, 3\to4, 4\to5$ forwards).

5. For each maturity $i \in \{1, 2, 3, 4, 5\}$, we form the daily log-return series

$$X_{i,j} = \log\!\left(\frac{r_{i,j+1}}{r_{i,j}}\right), \qquad j = 1, \ldots, 9,$$

where $r_{i,j}$ is the interpolated yield at maturity i on day j. We then compute the sample covariance matrix across maturities using these observations:

$$\Sigma_{\text{yield}} = \mathrm{Cov}(X_{\cdot,1}, \ldots, X_{\cdot,9}).$$

We repeat the same procedure for the four 1-year forward rates $(1\text{y}-1\text{y}), (1\text{y}-2\text{y}), (1\text{y}-3\text{y}), (1\text{y}-4\text{y})$ to obtain

$$\Sigma_{\text{fwd}}.$$

Numerical results are reported in reference (See Table 1 and Table 2).

6. Let Σ denote the covariance matrix (for yields or forwards). We compute

$$\Sigma v_k = \lambda_k v_k, \qquad \lambda_1 \geq \lambda_2 \geq \cdots.$$

The largest eigenvalue λ_1 measures the variance explained by the dominant factor (PC1), and its eigenvector v_1 describes the main co-movement pattern across maturities (e.g., parallel shift if loadings are similar, or twist/curvature if loadings have opposing signs).

Numerical results are reported in reference (See Table 3 and Table 4).

References

Github Link

https://github.com/EvelynHuang00/APM466A1

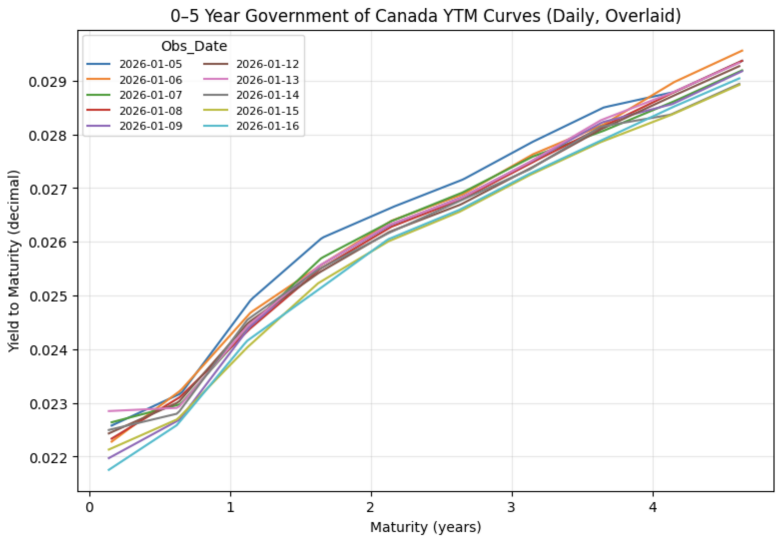

Figure 1: 0-5 year Government Canada YTM curves

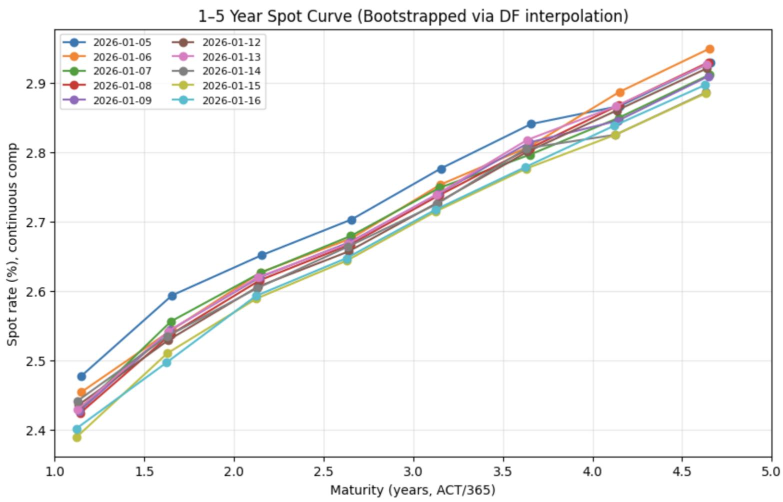

Figure 2: Bootstrapped 1–5Y Spot Curve

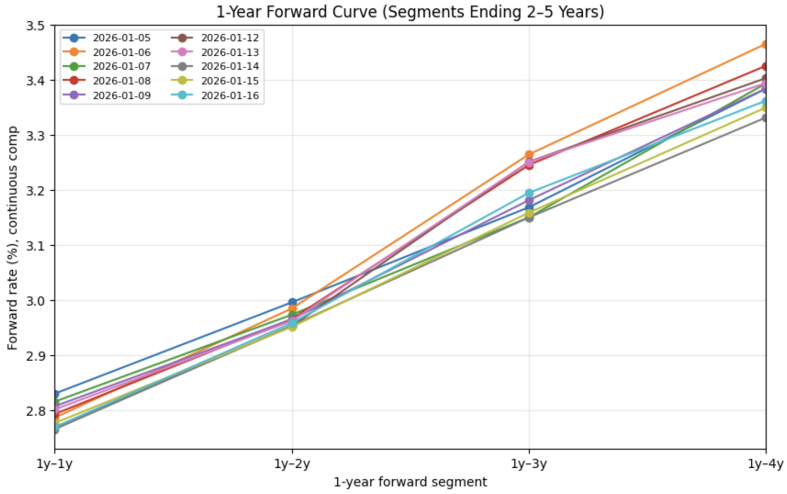

Figure 3: 1Y Forward Curve (2–5Y Segments)

	1Y	2Y	3Y	4Y	5Y
1Y	0.000055	0.000004	0.000003	0.000014	0.000011
2Y	0.000004	0.000028	0.000021	−0.000002	−0.000013
3Y	0.000003	0.000021	0.000017	0.000001	−0.000006
4Y	0.000014	−0.000002	0.000001	0.000037	0.000045
5Y	0.000011	−0.000013	−0.000006	0.000045	0.000061

Table 1: Covariance matrix of daily log-returns of yields (1Y–5Y).

	$1y1y$	$1y2y$	$1y3y$	$1y4y$
$1y1y$	0.000114	0.000017	−0.000134	−0.000074
$1y2y$	0.000017	0.000007	−0.000002	−0.000002
$1y3y$	−0.000134	−0.000002	0.000628	0.000343
$1y4y$	−0.000074	−0.000002	0.000343	0.000208

Table 2. Covariance matrix of daily log-returns of 1-year forward rates (1y1y–1y4y).

	PC1	PC2	PC3	PC4	PC5
λ	1.033498e-04	5.639675e-05	3.760027e-05	3.082759e-07	1.207623e-07
1Y	-0.314612	-0.778456	0.540449	0.040313	-0.036268
2Y	0.136963	-0.474968	-0.596816	-0.475917	-0.415885
3Y	0.065933	-0.344966	-0.495736	0.745570	0.273902
4Y	-0.574251	-0.012533	-0.278898	-0.385222	0.666257
5Y	-0.740371	0.221931	-0.167890	0.260013	-0.553899

Table 3. Eigenvalues and eigenvectors of the yield covariance matrix (sorted by descending eigenvalue).

	PC1	PC2	PC3	PC4
λ	0.000852	0.000086	0.000016	0.000004
1y1y	-0.203253	-0.959740	0.011351	0.193542
1y2y	-0.007023	-0.196554	-0.071300	-0.977872
1y3y	0.854478	-0.173760	-0.485342	0.064177
1y4y	0.478024	-0.100366	0.871338	-0.046792

Table 4. Eigenvalues and eigenvectors of the forward-rate covariance matrix (sorted by descending eigenvalue).